

1. $\min J = \int_{t_0}^{t_f} f dt$, t_0, t_f 给定, $x(t_0) = x_0, x(t_f) = x_f$, 求 x^* :
 必要条件: $\begin{cases} \frac{\partial f}{\partial x} - \frac{d}{dt}(\frac{\partial f}{\partial \dot{x}}) = 0 \rightarrow \text{欧拉方程} \\ x(t_0) = x_0, x(t_f) = x_f \end{cases}$ 无状态方程

2. 求 $x(t)$ 使满足 $x(t_0) = x_0$, 并使泛函 $J = \int_{t_0}^{t_f} f dt$ 最小 $x(t_f)$ 自由, t_f 给定
 可变端点问题

将 u 代入 f , 得 f 关于 $x, \dot{x}$ 的式子
 必要条件: $\begin{cases} \frac{\partial f}{\partial x} - \frac{d}{dt}(\frac{\partial f}{\partial \dot{x}}) = 0 \\ x(t_0) = x_0 \end{cases}$ 无状态方程
 $\frac{\partial f}{\partial \dot{x}}|_{t_f} = 0$

例: $\dot{x} = ax + u, x(0) = x_0$, 求 $u(t)$ s.t. $J = \frac{1}{2} \int_0^T (x^2 + r^2 u^2) dt$ 最小, t_f 固定, $x(t_f)$ 自由

解: $u = \dot{x} - ax$, 代入 f 得:

$$f = \frac{1}{2} x^2 + \frac{1}{2} r^2 (\dot{x}^2 - 2ax\dot{x} + a^2 x^2) = (\frac{1}{2} + \frac{1}{2} a^2 r^2) x^2 - ar^2 x\dot{x} + \frac{1}{2} r^2 \dot{x}^2$$

由 Euler 方程: $\frac{\partial f}{\partial x} - \frac{d}{dt}(\frac{\partial f}{\partial \dot{x}}) = 0$ 得:

$$\begin{cases} \ddot{x} = \frac{1}{r^2} (1 + a^2 r^2) x \\ x(0) = x_0 \\ (\dot{x} - ax)|_{t_f} = 0 \end{cases}$$

3. 有可动边界问题: t_f 也不固定

求 $x(t)$ 使 $x(t_0) = x_0$, $J(x) = \int_{t_0}^{t_f} f dt$ 极小 无状态方程

① $x(t_f)$ 与 t_f 相互独立: 必要条件为

$$\frac{\partial f}{\partial x} - \frac{d}{dt}(\frac{\partial f}{\partial \dot{x}}) = 0, x(t_0) = x_0, \frac{\partial f}{\partial \dot{x}}|_{t_f} = 0, f|_{t_f} = 0$$

② 终端点必须落到 $x = \varphi(t_f)$ 上: 必要条件为

$$\frac{\partial f}{\partial x} - \frac{d}{dt}(\frac{\partial f}{\partial \dot{x}}) = 0, x(t_0) = x_0, [f + (\dot{\varphi}(t_f) - \dot{x}) \frac{\partial f}{\partial \dot{x}}]|_{t_f} = 0$$

例: $l: x^2 = 1, x(0) = 1$, 求 $x(t)$ 使 $x(t)$ 落到 l 上, 并使 $J = \int_0^T \sqrt{1+x^2} dx$ 最小

3: l 与 x 与 t 相关连, $\therefore$

$$\text{Euler 方程: } \frac{d}{dt}(\frac{\partial f}{\partial \dot{x}}) = 0 \quad \therefore \frac{\dot{x}}{\sqrt{1+x^2}} = C \quad \therefore \dot{x}^2 = C^2 (1+x^2)$$

$\therefore$ 可得: $\dot{x} = \pm a, x = at + b$

$x(0) = b = 1$ 由边界条件:

$$\sqrt{1+x^2} + (-1-\dot{x}) \frac{x}{\sqrt{1+x^2}} = 0 \quad \therefore \dot{x}^2 + 1 = \dot{x}^2 + x^2$$

$\therefore \dot{x} = 1, x = t + b, x = b = 1 \quad \therefore x = t + 1$

4. 一般泛函: $J(x) = \theta(x(t_f), t_f) + \int_{t_0}^{t_f} f dt$ $x(t_0) = x_0, t_f$ 固定, $x(t_f)$ 自由

$$\text{必要条件: } \begin{cases} \frac{\partial f}{\partial x} - \frac{d}{dt}(\frac{\partial f}{\partial \dot{x}}) = 0 \\ x(t_0) = x_0, \frac{\partial f}{\partial \dot{x}}|_{t_f} = -\frac{\partial \theta(t_f)}{\partial x(t_f)} \end{cases}$$
 无状态方程
 $\frac{\partial f}{\partial \dot{x}}|_{t_f} = -\frac{\partial \theta(t_f)}{\partial x(t_f)}$

5. 无约束最优控制问题:

$$\dot{x} = f(x, u, t), x(t_0) = x_0, J(u) = \theta(x(t_f), t_f) + \int_{t_0}^{t_f} f(x, u, t) dt$$

由图3, $x(t)$ 自由, 求 u 使 $J(u)$ 极小

引入拉格朗日乘子 $\lambda(t)$, 构造哈密顿函数 $H = L + \lambda f$

必要条件: $\begin{cases} \frac{\partial H}{\partial u} = 0 \\ \dot{x} = f \\ x(t_0) = x_0 \\ \dot{\lambda} = -\frac{\partial H}{\partial x} \\ \lambda(t_f) = \frac{\partial \theta(x(t_f), t_f)}{\partial x(t_f)} \end{cases}$

当 t_f 也自由, 则增加边界条件: $H(t_f) + \frac{\partial \theta(t_f)}{\partial t_f} = 0$

$\lambda(t_f) = \frac{\partial \theta(t_f)}{\partial x(t_f)} \Rightarrow$ 当 $\lambda(t_f)$ 给定, 则换成 $x(t_f) = x_f$

步骤:

① 构造哈密顿函数 $H = L + \lambda f$, 并根据 $\frac{\partial H}{\partial u} = 0$ 求出 u^*

② 以 u^* 代入原方程, 并由必要条件解出 x^*, λ^*

③ 以 x^*, λ^* 代入 u^* , 得 u^* 最优控制

1. 当要求 $x(t_f)$ 满足约束 $N(x(t_f), t_f) = 0$, 且当 t_f 给定:

正则方程边界条件:

$\lambda(t_f) = \frac{\partial \theta(t_f)}{\partial x(t_f)} + \frac{\partial N^*(t_f)}{\partial x(t_f)} \cdot V$

$\begin{cases} x(t_0) = x_0 \\ \lambda(t_f) = \frac{\partial \theta(t_f)}{\partial x(t_f)} + \frac{\partial N(t_f)}{\partial x(t_f)} \cdot V \quad (V \text{ 为设的变量}) \\ N(t_f) = 0 \end{cases}$

$H(t_f) + \frac{\partial \theta(t_f)}{\partial t_f} + \frac{\partial N(t_f)}{\partial t_f} \cdot V = 0$

如果 t_f 也自由, 则再增加方程 $H(t_f) + \frac{\partial N(t_f)}{\partial t_f} \cdot V + \frac{\partial \theta(t_f)}{\partial t_f} = 0$

2. 最大值原理

设 $x = f, x(t_0) = x_0, u$ 有范围 U , 求 u s.t. $J = \theta(x(t_f), t_f) + \int_{t_0}^{t_f} L dt$ 最小

必要条件: $\begin{cases} \dot{x} = f \\ \dot{\lambda} = -\frac{\partial H}{\partial x} \end{cases}$

由于本类型 u 有限定范围, $\therefore$ 不写 $\frac{\partial H}{\partial u} = 0$, 其它类型均有 $\frac{\partial H}{\partial u} = 0$

根据变量范围取 u 使得 H 能到最小值

边界条件:

① 当 $x(t_f) = x_f$ 给定, 则: $x(t_0) = x_0, x(t_f) = x_f$

② 当 t_f 给定, $\lambda(t_f)$ 自由, 则: $x(t_0) = x_0, \lambda(t_f) = \frac{\partial \theta(t_f)}{\partial x(t_f)}$

③ 当 t_f 也自由: $x(t_0) = x_0, \lambda(t_f) = \frac{\partial \theta(t_f)}{\partial x(t_f)}, H(t_f) + \frac{\partial \theta(t_f)}{\partial t_f} = 0$

例题:

1. 对 $\dot{x} = -x + u, x(0) = 2$, 求 $|u(t)| \leq 1$, s.t. $J = \int_0^1 2x - u dt$ 取极小值

解: $-1 \leq u(t) \leq 1$

构造哈密顿函数: $H = 2x - u + \lambda(-x + u) = u(\lambda - 1) + (2 - \lambda)x$

由最大值原理, 应取 u 使 H 最小

$\therefore u(t) = \begin{cases} -1, & \lambda > 1 \\ \lambda - 1, & \lambda \leq 1 \end{cases}$

正则方程: $\begin{cases} \dot{x} = -x + u \\ \dot{\lambda} = -\frac{\partial H}{\partial x} = \lambda - 2 \end{cases}$

边界条件: $\begin{cases} x(0) = 2 \\ \lambda(1) = \frac{\partial \theta}{\partial x} = 0 \end{cases}$

$$|1, \lambda|$$

$$\therefore \lambda = 2 - 2e^{t-1} \quad \text{当 } \lambda = 1 \text{ 时, } t = -1.2 \approx 0.307$$

$$\therefore u(t) = \begin{cases} -1, & 0 \leq t < 0.307 \\ \text{不定}, & t = 0.307 \\ 1, & t > 0.307 \end{cases}$$

2. 给定 $\begin{cases} \dot{x}_1 = \lambda_2 & x_1(0) = 0 \\ \dot{x}_2 = -x_2^2 + u & x_2(0) = 0 \end{cases}$ 记 $x = (x_1, x_2)^T$, 求最优问题必要条件

① $t_f = 1$, $x(t_f)$ 自由, $J(u) = \frac{1}{2} \int_0^1 [(x_1 - 1)^2 + u^2] dt$

$$\therefore \text{令 } H = \frac{1}{2} x_1^2 - x_1 + \frac{1}{2} + \frac{1}{2} u^2 + \lambda_1 x_2 - \lambda_2 x_2^2 + \lambda_2 u$$

$$\frac{\partial H}{\partial u} = 0, \text{ 则: } u + \lambda_2 = 0, \therefore u = -\lambda_2$$

正则方程: $\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -x_2^2 + u \\ \dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = -x_1 + 1 \\ \dot{\lambda}_2 = -\frac{\partial H}{\partial x_2} = 2\lambda_2 x_2 - \lambda_1 \end{cases}$ 边界条件: $\begin{cases} x_1(0) = 0 \\ x_2(0) = 0 \\ \lambda_1(1) = \frac{\partial H(1)}{\partial x_1} = 0 \\ \lambda_2(1) = 0 \end{cases}$

② $t_f = 1$, $x(t_f) \in (1, 0)^T$, $|u(t)| \leq 1$, $J(u) = \frac{1}{2} \int_0^1 [(x_1 - 1)^2] dt$

$$\therefore \text{令 } H = \frac{1}{2} x_1^2 - x_1 + \frac{1}{2} + \lambda_1 x_2 - \lambda_2 x_2^2 + \lambda_2 u \quad (-1 \leq u(t) \leq 1)$$

$$\therefore u(t) = \begin{cases} 1, & \lambda_2 < 0 \\ \text{不定}, & \lambda_2 = 0 \\ -1, & \lambda_2 > 0 \end{cases} \quad \therefore u(t) = -\text{sgn}(\lambda_2)$$

正则方程: $\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -x_2^2 + u \\ \dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = -x_1 + 1 \\ \dot{\lambda}_2 = -\frac{\partial H}{\partial x_2} = 2\lambda_2 x_2 - \lambda_1 \end{cases}$ 边界条件: $\begin{cases} x_1(0) = 0 \\ x_2(0) = 0 \\ x(1) = 1 \\ x_2(1) = 0 \end{cases}$

③ $t_f = 1$, u^* 将 $x(0)$ 转移到 $(x_1 - 2)^2 + x_2^2 - 1 = 0$ 上, $J(u) = \frac{1}{2} \int_0^1 u^2 dt$

$$\therefore H = \frac{1}{2} u^2 + \lambda_1 x_2 - \lambda_2 x_2^2 + \lambda_2 u$$

$$\frac{\partial H}{\partial u} = 0 \Rightarrow u + \lambda_2 = 0$$

正则方程: $\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -x_2^2 + u \\ \dot{\lambda}_1 = 0 \\ \dot{\lambda}_2 = 2\lambda_2 x_2 - \lambda_1 \end{cases}$ 边界条件: $\begin{cases} x_1(0) = 0, x_2(0) = 0 \\ (x_1 - 2)^2 + x_2^2 - 1 = 0 \\ \frac{\partial H}{\partial x_1} + \frac{\partial V}{\partial x_1} \cdot V = 2(x_1(1) - 2)V = 0 \\ \frac{\partial H}{\partial x_2} + \frac{\partial V}{\partial x_2} \cdot V = 2x_2 V = 0 \end{cases}$

④ 求 u^* 将 $x(0)$ 转移到 $x(t_f) \in (1, 0)^T$, $J(u) = \frac{1}{2} \int_0^1 u^2 dt$

$$\therefore t_f = 1, H = \frac{1}{2} u^2 + \lambda_1 x_2 - \lambda_2 x_2^2 + \lambda_2 u \quad u + \lambda_2 = 0$$

正则方程: $\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -x_2^2 + u \end{cases}$ 边界条件: $\begin{cases} x_1(0) = 0 \\ x_2(0) = 0 \end{cases}$ 附加条件: $H(t_f) + \frac{\partial \theta(t_f)}{\partial t_f} = 0$
 $\therefore \frac{1}{2} u(1)^2 + \lambda_2(1) u(1) + 1 = 0$

$$\begin{cases} \lambda_1 = 0 \\ \dot{\lambda}_2 = 2\lambda_2 x_2 - \lambda_1 \end{cases}$$

$$\begin{cases} x_1(1) = 1 \\ x_2(1) = 0 \end{cases}$$

⑧ 求 u^* 满足 $|u(t)| \leq 1$, $x(0) \rightarrow x(t_f) = (1, 0)^T$, 且时间最短

3: $J = t_f = \int_0^{t_f} dt$, $\therefore L = 1$

$H = 1 + \lambda_1 x_2 - \lambda_2 x_2^2 + \lambda_0 u$, $-1 \leq u(t) \leq 1$ $t_f, x(t_f)$ 均不因 u

正则方程: $\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -x_2^2 + u \\ \dot{\lambda}_1 = 0 \\ \dot{\lambda}_2 = 2\lambda_2 x_2 - \lambda_1 \end{cases}$ 边界条件: $\begin{cases} x_1(0) > 0 \\ x_2(0) > 0 \\ x_1(t_f) = 1 \\ x_2(t_f) = 0 \end{cases}$

附加条件: $H(t_f) + \frac{\partial H(t_f)}{\partial t_f} = 0$, 即: $1 + \lambda_2(t_f)u(t_f) = 0$

⑨ $t_f = 1$, $|u(t)| \leq 1$, $x(0) \rightarrow x(t_f) = (1, 0)^T$, $J(u) = \int_0^1 |u| dt$

3: $-1 \leq u(t) \leq 1$

$H = |u| + \lambda_1 x_2 - \lambda_2 x_2^2 + \lambda_0 u$

$\therefore (I) 0 \leq u(t) \leq 1$, 则: $H = u(t)\lambda_0 + \lambda_1 x_2 - \lambda_2 x_2^2$

$\therefore u(t) = \begin{cases} 1, & \lambda_2 \leq -1 \\ 0, & \lambda_2 > -1 \end{cases}$ 正则方程、边界条件不变

(II) $-1 \leq u(t) \leq 0$, 则: $H = (\lambda_2 - 1)u + \lambda_1 x_2 - \lambda_2 x_2^2$

$\therefore u(t) = \begin{cases} -1, & \lambda_2 > 1 \\ 0, & \lambda_2 \leq 1 \end{cases}$